

Example: Calcite in equilibrium

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1 Theory

We solve the following system of equations:

$$\begin{cases} \phi \frac{\partial \mathbf{c}}{\partial t} = \mathbf{M} [-\mathbf{q} \cdot \nabla \mathbf{c} + \nabla \cdot (\mathbf{D} \nabla \mathbf{c})] + \phi \mathbf{S}_e^T \mathbf{r}_e \\ \mathbf{S}_e \log \mathbf{a} = \log \mathbf{K}_e, \end{cases} \quad (1)$$

where $\mathbf{c}(\mathbf{x}, t)$ is the vector of concentrations of all species (of dimension n_{sp} , number of species); $\phi \in (0, 1]$ is porosity; $\mathbf{M} \in \mathbb{R}^{n_{sp} \times n_{sp}}$ is a diagonal matrix such that $M_{ii} = 1$ if the species is mobile and 0 otherwise; $\mathbf{q}$ is the Darcy flux, $\mathbf{D}$ is the hydrodynamic dispersion tensor (scalar in 1D); $\mathbf{a} \in \mathbb{R}^{n_{sp}}$ is the activity of all species; $\mathbf{S}_e \in \mathbb{R}^{n_e \times n_{sp}}$ is the stoichiometric matrix of equilibrium reactions, whose reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ (n_e and are the number of equilibrium reactions) and $\mathbf{K}_e$ are the equilibrium constants, modified as proposed by Wang et al. (2025) so as to account for activity, temperature and water mass fraction. The last term in (1) is the contribution of chemical reactions to the mass balance of species. The boundary conditions for every species i are prescribed mass flux at the inflow

$$-M_{ii} (-\mathbf{q} c_{i,1} + \mathbf{D} \nabla c_{i,1}) \cdot \mathbf{n} = M_{ii} \mathbf{q}_{inf} c_{i,inf} \cdot \mathbf{n}$$

and no dispersive flux at the outflow

$$c_{i,n} = c_{i,out} \quad (3)$$

where we have multiplied by M_{ii} to highlight that boundary conditions only apply to mobile species, since (1) is local for immobile species. More details on how to impose different boundary conditions in the ADE can be found at Petchamé-Guerrero and Carrera (2024).

The number of equations in (1)-(2) is $n_{sp} + n_e + n_K$. We must solve these equations for each target (cell, node, or particle depending on the solution method), therefore, the total number of equations is $n \times (n_{sp} + n_e + n_K)$, where n is the number of targets.

For convenience, we write concentrations and reaction rates in matrix form, where columns are targets and rows are species and reactions, respectively. Let us denote $\mathbf{c}_{ext} = (\mathbf{c} \mid \mathbf{c}_e) \in \mathbb{R}^{n_{sp} \times (n+n_e)}$ the concentrations matrix comprising both $\mathbf{c} \in \mathbb{R}^{n_{sp} \times n}$, matrix of concentrations at all targets, and $\mathbf{c}_e \in \mathbb{R}^{n_{sp} \times n_e}$, matrix of concentrations of external waters (those entering the model domain associated to water sources and those at the boundaries). Let us also denote $\mathbf{A}_{ext} = (\mathbf{A} \mid \mathbf{E}) \in \mathbb{R}^{n \times (n+n_e)}$ the extended transport matrix comprising both the transport matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and the external matrix $\mathbf{E} \in \mathbb{R}^{n \times n_e}$ of external waters contributions to the solute mass balance. The discretised version of (1) becomes

$$\frac{\mathbf{c}^{k+1} - \mathbf{c}^k}{\Delta t^k} \mathbf{F} = \mathbf{M} \left[\mathbf{c}_{ext} (\mathbf{A}_{ext})^T \right]^{k+\theta_t} + \mathbf{S}_e^T \mathbf{r}_e^{k+\theta_r} \mathbf{F} \quad (4)$$

where $\mathbf{F} \in \mathbb{R}^{n \times n}$ is the water storage matrix, which is diagonal if and only if the mesh is homogeneous ($F_{ii} = \phi_i$ in differential formulations such as FD, $F_{ii} = V_i \phi_i$ in integral formulations such as lumped FEM or FV); Δt^k is the time increment; $\theta_t \in (0, 1]$ and $\theta_r \in (0, 1]$ represent the time weighting factor for transport and reactions, respectively; the variables with superscript $k + \theta_t$ are evaluated at some time between time steps k and $k + 1$, that is, $\mathbf{c}_{ext}^{k+\theta_t} = \theta_t \mathbf{c}_{ext}^{k+1} + (1 - \theta_t) \mathbf{c}_{ext}^k$ (idem for θ_r). To preserve mass balance, the rows of matrix $\mathbf{A}_{ext}^{k+\theta_t}$ must add up to 0, because in the absence of reactions, the total mass remains unchanged. Once we transpose the matrix, it is the columns that must have zero sum.

Matrix $\mathbf{A}_{ext}$ depends on the transport formulation. Here, we adopt the advective Eulerian form of the ADE proposed by Petchamé-Guerrero and Carrera (2024).

We use the method of components (De Simoni et al., 2005) to eliminate the equilibrium reactions from the reactive transport equation, therefore we obtain a conservative transport equation:

$$\frac{\mathbf{u}^{k+1} - \mathbf{u}^k}{\Delta t^k} \mathbf{F} = \left[\mathbf{u}_{ext} (\mathbf{A}_{ext})^T \right]^{k+\theta_t} \quad (5)$$

where $\mathbf{u}_{ext} = (\mathbf{u} \mid \mathbf{u}_e) \in \mathbb{R}^{n_p \times (n+n_e)}$ are the extended concentrations of components, defined as $\mathbf{u}_{ext} = \mathbf{U}\mathbf{c}_{ext,nc}$ (the subscript nc represents the subset of variable activity species, and the subscript p the subset of primary species).

The WMA formula (Soler-Sagarra et al., 2022) for computing the component concentrations in the whole mesh in the time step $k+1$ results from rearranging equation (5), which produces

$$\mathbf{u}^{k+1} = \mathbf{u}_{ext}^k \mathbf{\Lambda}_{ext}^k + \mathbf{u}_e^{k+1} \mathbf{\Lambda}_e^{k+1} \quad (6)$$

where $\mathbf{B}^{k+1} = (\mathbf{I}_n - \Delta t^k \theta_t (\mathbf{A}^{k+1})^T)^{-1}$, $\mathbf{\Lambda}_{ext}^k = \begin{pmatrix} \mathbf{I}_n + \Delta t^k (1 - \theta_t) (\mathbf{A}^k)^T \\ \Delta t^k (1 - \theta_t) (\mathbf{E}^k)^T \end{pmatrix} \mathbf{F}^{-1} \mathbf{B}^{k+1}$ and $\mathbf{\Lambda}_e^{k+1} = \Delta t^k \theta_t (\mathbf{E}^{k+1})^T \mathbf{F}^{-1} \mathbf{B}^{k+1}$.

The matrices $\mathbf{\Lambda}_{ext}^k$ and $\mathbf{\Lambda}_e^{k+1}$ can be interpreted as containing the mixing ratios of all the waters that mix with each other between time steps k and $k+1$. Therefore, the r.h.s. of equation (6) represent conservative mixing. Moreover, it becomes more elegant if external concentrations are stationary, which is the case in this example. In this case, $\mathbf{u}_e^{k+1} = \mathbf{u}_e^k$ and therefore we have the simplified WMA equation

$$\mathbf{u}^{k+1} = \mathbf{u}_{ext}^k \mathbf{\Lambda}_{ext}^k \quad (7)$$

with $\mathbf{\Lambda}_{ext}^k = \begin{pmatrix} \mathbf{I}_n + \Delta t^k (1 - \theta_t) (\mathbf{A}^k)^T \\ \Delta t^k (\mathbf{E}^{k+\theta_t})^T \end{pmatrix} \mathbf{F}^{-1} \mathbf{B}^{k+1}$.

The mixing ratios will depend on the transport time weighting factor θ_t . The mixing ratios "pseudomatrices" might have columns with different dimensions, depending on the amount of waters that mix with each other at each target water, hence the use of the prefix "pseudo". It is important to point out that the mixing ratios apply to domain waters as well as boundary or recharge waters, therefore we use the extended matrix $\mathbf{c}_{ext}$ for the concentrations of aqueous species in all of these different waters.

The WMA scheme (7) applied to each target j gives the following system of equations to be solved at every time step $k+1$:

$$\mathbf{u}_j^{k+1} = \sum_{i \in I_j} \lambda_{ij}^k \mathbf{u}_{ext,i}^k \quad (8)$$

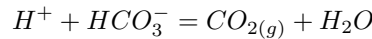
where I_j is the set of waters that mix with water j (including itself) and λ_{ij}^k are the mixing ratios. At each time step and target, we must speciate to compute the variable activity species concentrations from the component concentrations (see Appendix A for the details).

2 Chemistry

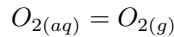
We consider a simplified reactive transport problem with two waters: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite $CaCO_3$. The aqueous species involved are H_2O , H^+ , Ca^{2+} , $O_{2(aq)}$ and HCO_3^- . The concentrations are written in Table 1.

There are no kinetic reactions in the chemical system. The (equilibrium) reactions in the inflow water are

- Atmospheric CO_2 :

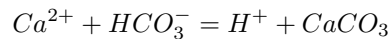


- Atmospheric O_2 :



and in the resident water is

- Calcite:



The stoichiometric matrix at the inflow is

$$\mathbf{S}_e^{inf} = \begin{pmatrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- & H_2O & O_{2(g)} & CO_{2(g)} \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

64 since there are no mineral reactions. In the domain, there is only one equilibrium reaction (calcite), therefore the stoichiometric
 65 matrix is

$$\mathbf{S}_e^{dom} = \begin{pmatrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- & H_2O & CaCO_3 \\ 1 & -1 & 0 & -1 & 0 & 1 \end{pmatrix}$$

66 There are 3 primary species in the calcite mineral zone: H^+ , Ca^{2+} and $O_{2(aq)}$; therefore, the component matrix is

$$\mathbf{U} = (\mathbf{I} \mid \mathbf{U}_2)$$

67 where

$$\mathbf{U}_2 = \begin{pmatrix} HCO_3^- \\ 1 \\ -1 \\ 0 \end{pmatrix}.$$

	inflow	resident
H^+	10^{-7}	10^{-5}
Ca^{2+}	10^{-9}	10^{-3}
$O_{2(aq)}$	$2.52947 \cdot 10^{-4}$	10^{-6}
HCO_3^-	$5.37611 \cdot 10^{-5}$	$7.05830 \cdot 10^{-1}$

Table 1: Aqueous species concentrations of inflow and resident waters (in molalities)

68 3 Transport

69 We consider a 1D Eulerian mesh with $n = 10$ cells and cell size $\Delta x = 0.01m$. We use a time step $\Delta t = 0.01d$. We solve
 70 transport fully implicitly, that is, $\theta_t = 1$. The transport properties are $\phi = 0.5$, $q = 0.2m/d$, $\alpha_L = 0.5m$ where ϕ is porosity, q
 71 is Darcy's flow and α_L is longitudinal dispersivity. Dispersion $D = \alpha_L q$ is therefore $D = 0.01m^2/d$. We assume that the inflow
 72 flux is the same as in the mesh. The transport matrix $\mathbf{A}$ has the following values

$$\forall i \in \{2, \dots, n-1\}, \quad A_{ij} = \begin{cases} \frac{D}{\Delta x^2} - \frac{q}{2\Delta x} & j = i+1 \\ \frac{D}{\Delta x^2} + \frac{q}{2\Delta x} & j = i-1 \\ -\frac{2D}{\Delta x^2} & j = i \\ 0 & \text{otherwise} \end{cases}$$

73 The boundary conditions are

$$A_{ij} = \begin{cases} -\frac{q}{2\Delta x} - \frac{D}{\Delta x^2} & i = j = 1 \\ \frac{D}{\Delta x^2} - \frac{q}{2\Delta x} & i = 1, \quad j = 2 \\ \frac{q}{2\Delta x} + \frac{D}{\Delta x^2} & i = n, \quad j = n-1 \\ -\frac{q}{2\Delta x} - \frac{D}{\Delta x^2} & i = j = n \end{cases}$$

74

$$E_{i,1} = \begin{cases} \frac{q}{\Delta x} & i = 1 \\ 0 & i \neq 1 \end{cases}$$

75 where $\mathbf{E}$ is the external matrix.

Appendices

A Speciation from components

Once we have computed $\mathbf{u}_j^{k+1}$ explicitly, that is, component concentrations at target j and time step $k + 1$, we must speciate to compute the variable activity species concentrations. Let $\mathbf{U}_j := (\mathbf{U}_1 \mid \mathbf{U}_{2,nc})$. This involves solving for each target water j the nonlinear system of equations

$$\begin{cases} f_k(\mathbf{c}_{1_j}^{k+1}) = \mathbf{U}_1 \mathbf{c}_{1_j}^{k+1} + \mathbf{U}_{2,nc} \mathbf{c}_{2,nc_j}^{k+1} - \mathbf{u}_j^{k+1} \\ \log_{10} \mathbf{c}_{2,nc_j}^{k+1} = \mathbf{S}_{e,nc_1}^* \log_{10}(\gamma_{1_j}^{k+1} \mathbf{c}_{1_j}^{k+1}) + \log_{10} \mathbf{K}_e^* - \log_{10} \gamma_{2,nc_j}^{k+1} \end{cases} \quad (9)$$

where the subscript 1 accounts for primary species and 2, nc for secondary variable activity species. The matrices $\mathbf{S}_{e,nc_1}^*$ and $\mathbf{K}_e^*$ are defined as

$$\begin{cases} \mathbf{S}_{e,nc_1}^* = -\mathbf{S}_{e,nc_2}^{-1} \mathbf{S}_{e,1} \\ \log_{10} \mathbf{K}_e^* = -\mathbf{S}_{e,nc_2}^{-1} \log_{10} \mathbf{K}_e. \end{cases}$$

We use Newton-Raphson, where f_k is the residual function. The Jacobian is therefore

$$\frac{\partial f_k(\mathbf{c}_{1_j}^{k+1})}{\partial \mathbf{c}_{1_j}^{k+1}} = \mathbf{U}_1 + \mathbf{U}_{2,nc} \frac{\partial \mathbf{c}_{2,nc_j}^{k+1}}{\partial \mathbf{c}_{1_j}^{k+1}}. \quad (11)$$

To compute $\frac{\partial \mathbf{c}_{2,nc_j}^{k+1}}{\partial \mathbf{c}_{1_j}^{k+1}}$ we must solve the linear system for the logarithms:

$$\left(\mathbf{I} - \mathbf{S}_{e,nc_1}^* \frac{\partial \log_{10} \gamma_{1_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{2,nc_j}^{k+1}} + \frac{\partial \log_{10} \gamma_{2,nc_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{2,nc_j}^{k+1}} \right) \frac{\partial \log_{10} \mathbf{c}_{2,nc_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{1_j}^{k+1}} = \mathbf{S}_{e,nc_1}^* \left(\frac{\partial \log_{10} \gamma_{1_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{1_j}^{k+1}} + \mathbf{I} \right) - \frac{\partial \log_{10} \gamma_{2,nc_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{1_j}^{k+1}} \quad (12)$$

In the case of constant activity coefficients, it becomes much simpler, since

$$\frac{\partial \log_{10} \mathbf{c}_{2,nc_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{1_j}^{k+1}} = \mathbf{S}_{e,nc_1}^*.$$

Then, by definition of logarithm, we get

$$\frac{\partial \mathbf{c}_{2,nc_j}^{k+1}}{\partial \mathbf{c}_{1_j}^{k+1}} = \text{diag}(\mathbf{c}_{2,nc_j}^{k+1}) \frac{\partial \log_{10} \mathbf{c}_{2,nc_j}^{k+1}}{\partial \log_{10} \mathbf{c}_{1_j}^{k+1}} \text{diag}\left(\frac{1}{\mathbf{c}_{1_j}^{k+1}}\right) \quad (13)$$

Let ε_a be the absolute tolerance and ε_r be the relative tolerance. Let us impose a maximum number of iterations $n_{it,max}$. The algorithm for every time step $k + 1$ and target water j is

1. Compute $\mathbf{u}_j^{k+1}$ using (8).
2. Set $\mu = 0$.
3. Set $i = 0$ and initialize concentration primary species:

$$\left(\mathbf{c}_{1_j}^{k+1} \right)^i = (1 + \mu) \mathbf{c}_{1_j}^k - \mu \mathbf{c}_{1_j}^{k-1}, \quad 0 \leq \mu \leq 1 \quad (14)$$

4. Given $\left(\mathbf{c}_{1_j}^{k+1} \right)^i$, compute $\left(\mathbf{c}_{2,nc_j}^{k+1} \right)^i$ using (10) with the fixed-point method
5. Compute $f_k\left(\left(\mathbf{c}_{1_j}^{k+1}\right)^i\right)$ using (9). If $\left\| f_k\left(\left(\mathbf{c}_{1_j}^{k+1}\right)^i\right) \right\| < \varepsilon_a$ then convergence is attained:

- (a) Set $\mathbf{c}_{nc_j}^{k+1} = \left(\mathbf{c}_{nc_j}^{k+1} \right)^i$
- (b) Exit algorithm.

6. Compute $\frac{\partial \left(\mathbf{c}_{2,nc_j}^{k+1} \right)^i}{\partial \left(\mathbf{c}_{1_j}^{k+1} \right)^i}$ using (13)

7. Compute $\frac{\partial f_k \left(\left(\mathbf{c}_{1_j}^{k+1} \right)^i \right)}{\partial \left(\mathbf{c}_{1_j}^{k+1} \right)^i}$ using (11).

8. Solve linear system

$$\frac{\partial f_k \left(\left(\mathbf{c}_{1_j}^{k+1} \right)^i \right)}{\partial \left(\mathbf{c}_{1_j}^{k+1} \right)^i} \Delta \left(\mathbf{c}_{1_j}^{k+1} \right)^i = -f_k \left(\left(\mathbf{c}_{1_j}^{k+1} \right)^i \right)$$

where $\Delta \left(\mathbf{c}_{1_j}^{k+1} \right)^i = \left(\mathbf{c}_{1_j}^{k+1} \right)^{i+1} - \left(\mathbf{c}_{1_j}^{k+1} \right)^i$.

9. If

$$\left\| \frac{\Delta \left(\mathbf{c}_{1_j}^{k+1} \right)^i}{\left(\mathbf{c}_{1_j}^{k+1} \right)^i} \right\| < \varepsilon_r$$

then go to step 2 and increase μ . If $\mu = 1$, algorithm does not converge. You must either increase tolerance and therefore restart the time loop or change initialisation formula (14).

10. Update $\mathbf{c}_{1_j}^{k+1}$ with control factor $\alpha \in (0, 1)$:

$$\left(\mathbf{c}_{1_j}^{k+1} \right)^{i+1} = \left(\mathbf{c}_{1_j}^{k+1} \right)^i + \Delta \left(\mathbf{c}_{1_j}^{k+1} \right)^i$$

$$\left(\mathbf{c}_{1_j}^{k+1} \right)^{i+1} = \begin{cases} \alpha \left(\mathbf{c}_{1_j}^{k+1} \right)^i & \text{if } \left(\mathbf{c}_{1_j}^{k+1} \right)^{i+1} \leq \alpha \left(\mathbf{c}_{1_j}^{k+1} \right)^i \\ \frac{\left(\mathbf{c}_{1_j}^{k+1} \right)^i}{\alpha} & \text{if } \left(\mathbf{c}_{1_j}^{k+1} \right)^{i+1} \geq \frac{\left(\mathbf{c}_{1_j}^{k+1} \right)^i}{\alpha} \end{cases}$$

11. Set $i = i + 1$. If $i > n_{it,max}$ then the algorithm does not converge fast enough. Follow the same procedure as in step 9.

12. Go to step 4.

References

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